

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (CE/IT/EC) Examination

April-May, 2013

CL101.01 Fundamentals of Civil Engineering

Date: 07.05.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Q-3 must be attempted in answer sheet only.

SECTION – I

- Q - 1 (a) (i) Enlist qualities of good cement. [06]
(ii) Explain in detail the ingredients for Reinforced Cement Concrete.
- (b) Which branch of civil engineering deals with the waste water treatment and solid waste management? [01]
- Q - 2 (a) Define regulatory signs and warning signs. Draw sketches of different regulatory signs and warning signs. [06]
- (b) (i) Enlist the branches of transportation engineering. [04]
(ii) Write short note on structural engineering.
- (c) Give your suggestions for good orientation of building. [04]

OR

- Q - 2 (a) Write detail short note on BOT projects. [06]
(b) Summarize the main duties of civil engineers. [04]
(c) Write different elements of a framed structure from bottom to top. [04]
- Q - 3 **Figure-1** shows line diagram of a plan. Draw wall thickness plan of the same. Use data given with the figure. Show all the necessary details and assume suitable scale. [14]

OR

- Q - 3 **Figure-1** shows line diagram of a plan. Draw elevation from the face, where arrow is given. Use levels as given with the figure and also provide windows and doors at suitable positions. Assume suitable scale. [14]

SECTION – II

- Q - 4 (a) Explain water storage systems and water conveyance systems, in detail. [05]
(b) Define **any two**; (i) Evaporation, (ii) Runoff, (iii) infiltration. [02]
- Q - 5 (a) Following perpendicular offsets were taken from a chain line at an interval of 10m to a curved boundary. Calculate the area by average ordinate method and trapezoidal

rule;

0, 7.830, 5.260, 6.450, 7.330, 7.870, 8.230 and 0.

- (b) (1) Convert following whole circle bearings to quadrantal bearings; [05]
(i) $25^{\circ}30'$ (ii) 135° (iii) $281^{\circ}30'$
(2) Convert following quadrantal bearings to whole circle bearings;
(i) $N45^{\circ}E$ (ii) $S89^{\circ}W$
- (c) A 30m chain was found to be 0.15m too long after chaining a distance of 5000m. It [04]
was found to be 0.30m too long after measuring a total distance of 10000m. At the
start of the work, the chain was tested and was found to be exactly 30m in length.
Find out the correct length of the measured distance.

OR

- Q - 5 (a) Following readings were obtained when the perimeter of a rectangle 35cm X 20cm [05]
was traversed clockwise by a planimeter with the anchor point inside;

Initial Reading = 1.349 & Final Reading = 9.854

The zero of the dial passed the fixed index mark twice in the anti-clockwise direction.

Calculate the area of zero-circle. Take $M = 100\text{cm}^2$.

- (b) The following bearings were observed in running a closed traverse ABCDE. [05]
Calculate the interior angles of the traverse and apply necessary checks.

Line	Fore Bearing
AB	90°
BC	225°
CD	135°
DE	270°
EF	315°
FA	45°

- (c) Enlist the instruments required for chain surveying. Draw neat sketch of any two [04]
instruments and their functions.

- Q - 6 (a) Briefly explain the temporary adjustment of dumpy level, over the station point. [06]

- (b) Define the following terms; Axis of telescope, Fore sight, Mean sea level, Bench [05]
mark, Height of instrument.

- (c) Define following; [03]
Hydrology, Hydrological cycle, Interception

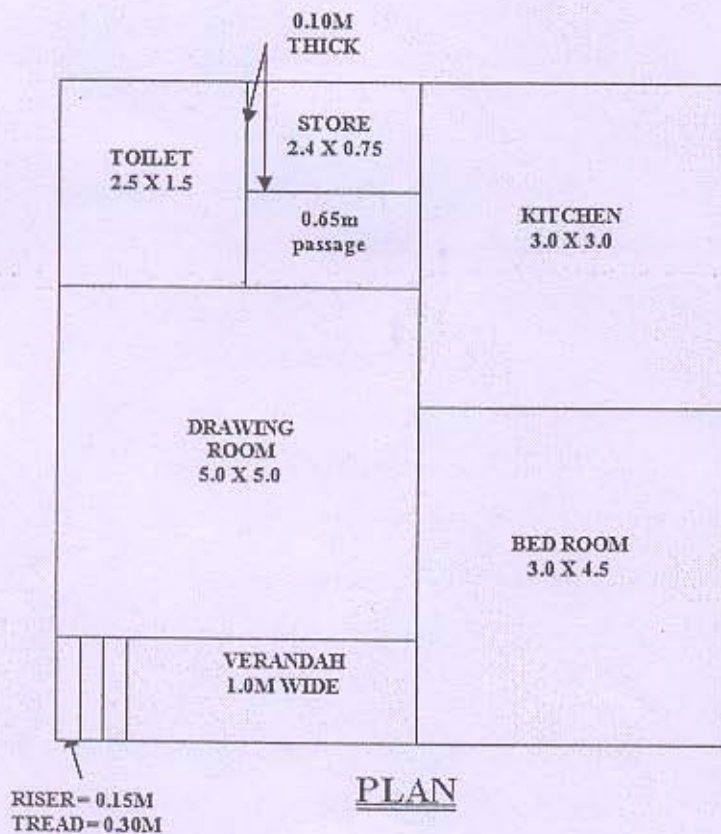
OR

- Q - 6 (a) The following consecutive readings were taken with a dumpy level: 1.345, 1.830, [06]
2.165, 2.765, 0.325, 1.140, 2.125, 2.890, 3.320, 0.990 and 1.425

The instrument was shifted after fourth and ninth readings. The first reading was taken on Bench mark of 250.000m RL. Use Height of Instrument method to calculate the reduced levels of all points and apply usual checks.

(b) Enlist the types of errors occur during compass survey. Explain any two in detail. [05]

(c) Explain any three water losses, in detail. [03]



PLAN

REQUIRED LEVELS:-

Plinth Level = 0.60 m above Ground Level

Lintel Level = 2.00 m above Plinth Level

Slab Bottom = 3.00 m above Plinth Level

Slab Thickness = 0.10 m thick

Parapet Height = 0.90 m above Slab Top.

Note: All walls are 30 cm thick; accept where mentioned in above figure.

Figure-1